
Auditing VLM Benchmarks from Existing Evaluation Runs

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Abstract

1 Every evaluation run of a vision-language model (VLM) already produces per-
2 question artifacts (predictions, judged correctness, metadata) that leaderboards
3 reduce to a single score and then discard. We reuse those artifacts to audit the
4 benchmark itself, and add an audit layer to VLMEvalKit: six detectors that run
5 as post-processing over saved outputs, plus one additional text-only rerun with
6 visual inputs removed, emitting a per-question report in a single command. The
7 layer reports per-question findings together with aggregate audit statistics; for the
8 headline quantities, it attaches permutation-null excess where applicable, interval
9 estimates for proportions, and sensitivity to the model pool. Auditing MMStar, a
10 benchmark curated specifically so that the image is indispensable, with three mod-
11 els, we find zero aggregate accuracy gain from the image on 20.2% of its questions.
12 An item-matched permutation null accounts for 17.0%, leaving a +3.2 pp excess
13 (corrected permutation $p = 0.001$). This excess is concentrated in the mathe-
14 matics and science categories and is absent from coarse perception, a localization
15 consistent with knowledge substituting for visual evidence; confirming that mech-
16 anism would require item-level review. A second detector flags 134 questions
17 on which every model that answered in parseable form picked the same non-gold
18 option, and these candidates peak in the perception categories, whereas the zero-
19 gain excess is largest in mathematics and science & technology. The contrasting
20 category profiles show that the two detectors capture distinct signals. Replacing
21 the LLM judge with exact string matching changes no conclusion, and every scor-
22 ing discrepancy between the two traces to one model whose free-form answers a
23 parser cannot read. On a binary yes/no benchmark the detectors measurably fall
24 out of calibration, a scope boundary that we measure and report. The audit layer,
25 the prediction artifacts and the generated reports accompany this submission.

26 1 Introduction

27 MMStar was built by filtering candidate questions so that the image is essential: items solvable
28 by blind language models were removed by construction. We run three VLMs on it twice, first as
29 intended and then with every image removed from the prompt, and align the two runs question by
30 question. On 303 of 1,500 questions (20.2%) the pool is exactly as accurate without the image as
31 with it. Most of that share is plain arithmetic: even blind answers unrelated to the items sometimes
32 tie full-run accuracy, and an item-matched permutation null puts that baseline at 17.0%. The remain-
33 ing +3.2 pp ($p = 0.001$) is a paired zero-gain signal, and it concentrates in the two categories where
34 non-visual knowledge is a plausible explanation. Finding this required no new benchmark and no
35 annotation campaign. The audit reuses the full-run artifacts and adds one text-only inference pass
36 per model; this pass avoids visual processing but is not cost-free.

That observation motivates this paper. Multimodal benchmarks are known to contain items solvable without the image, wrong gold labels, position and selection biases in multiple choice, and scores that shift with the answer-extraction protocol. The community usually responds by building cleaner replacement benchmarks or by running dedicated flaw-mining pipelines, and both responses treat auditing as a separate research project with its own inference budget. We take an infrastructure view: the per-question artifacts that evaluation toolkits persist on every run are already sufficient to audit the benchmark as a side effect of evaluating on it. We implement this view as an audit layer inside a mainstream evaluation toolkit and validate it in a case study designed to probe its own blind spots as well as the benchmark’s.

1.1 Contributions

1. **An audit layer built into an evaluation toolkit.** Six benchmark-pathology detectors consume the per-question files VLMEvalKit already writes; the only added inference is a text-only rerun, implemented in the standard local and API wrapper paths by removing visual content before generation (Section 3).
2. **A statistical reading of detector output.** A raw flag count is hard to interpret, because part of it is produced by marginal accuracies alone. We report the excess over an item-matched permutation null, with interval estimates, leave-one-model-out ranges and per-category strata, and show that the excess is far more pool-stable than the raw share (Section 3.3).
3. **A case study that localizes two audit signals and measures the audit’s own limits.** On MMStar, an excess zero-gain rate and unanimous label-conflict candidates are recovered from the same artifacts and show different category profiles. The same artifacts quantify how much the audit depends on an LLM judge, trace the residual dependence to a single model’s output format, and establish a measured scope boundary on binary yes/no items (Sections 4.1–4.5).

The pathologies these detectors target are documented at scale by prior and concurrent work (Section 2). The contribution of this paper is architectural and statistical: the checks run inside the evaluation infrastructure, on artifacts it already saves, with uncertainty attached, so that every evaluation run can double as a benchmark audit.

2 Related work

Benchmark pathologies. Language priors in visual question answering were established by Goyal et al. [2017]; MMStar [Chen et al., 2024] and MMEvalPro [Huang et al., 2024] showed that blind language models score far above chance on multimodal multiple-choice suites; concurrent work probes the visual dependence of existing suites through blur ablations [Zhou et al., 2026] or phrasing-controlled items [Singla et al., 2026]. Label errors destabilize rankings [Northcutt et al., 2021]; re-annotation efforts include MMLU-Redux [Gema et al., 2025] and platinum benchmarks [Vendrow et al., 2025], and Li et al. [2025] report a fallibility analysis covering MMStar. Position and selection bias in multiple choice is established for language models [Zheng et al., 2024], handled in MMBench through CircularEval [Liu et al., 2024], and reused as a contamination probe [Song et al., 2024]. Kim et al. [2025] show that model errors correlate across vendors and scales, which caps what any cross-model consensus can establish; we return to this in Sections 4.3 and 5.

Audit pipelines and toolkits. The standard responses to these pathologies treat auditing as a separate research project with its own inference budget, either by building cleaner replacement benchmarks [Chen et al., 2024, Li et al., 2024, Vendrow et al., 2025, Joshi et al., 2026] or by running standalone flaw-mining pipelines. Fantastic Bugs [Truong et al., 2025] mines response patterns to flag invalid questions in nine text benchmarks; BenchMarker [Balepur et al., 2026] applies an education-inspired rubric with LLM judges to text multiple-choice benchmarks; DatBench [Joshi et al., 2026] audits 33 VLM datasets and releases a cleaned suite. Our deliverable differs: audit machinery embedded in a mainstream toolkit, reusable on any benchmark-model combination from artifacts the toolkit already saves. To our knowledge no such layer exists for VLMEvalKit [Duan et al., 2024] or LMMs-Eval [Zhang et al., 2024]. Blind-solvability and label errors in VLM benchmarks are documented at scale by concurrent work [Joshi et al., 2026, Zhou et al., 2026]; the checks themselves are therefore established, and this paper moves them inside the evaluation infrastructure.

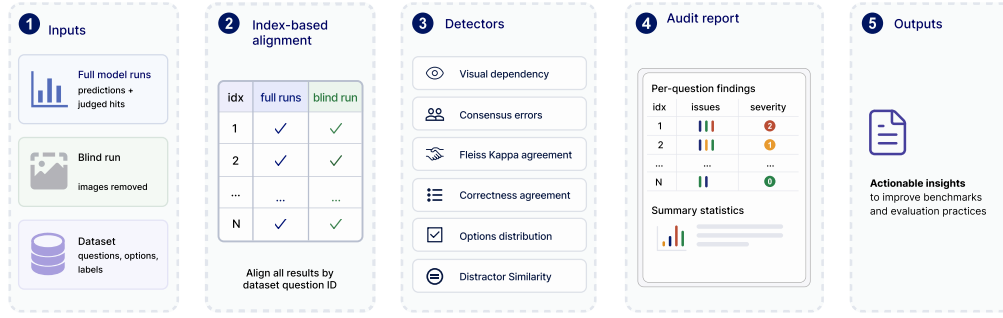


Figure 1: The audit pipeline in five stages. **(1) Inputs:** the full model runs with predictions and judged hits, one additional text-only blind run with images removed, and the dataset itself with questions, options and gold labels. **(2)** All frames are aligned by dataset question ID. **(3)** Six detectors process the aligned frame. **(4)** The audit report lists per-question findings with severity levels and summary statistics, including permutation-null excess values and a reproducibility manifest (git commit, command-line arguments, generation parameters). **(5)** The report yields actionable insights for improving benchmarks and evaluation practice. The blind run is the only step that requires inference, and it runs without images.

Redundancy analyses over existing score matrices [Zhang et al., 2025] share the reuse-the-exhaust spirit for a different question. On the measurement side, Rosenthal et al. [2025] document protocol sensitivity of VQA accuracy; we extend that question from accuracy numbers to audit conclusions.

3 Methodology

3.1 Inputs and the blind-run protocol

The layer consumes the per-question result files of n models on one benchmark, each with a raw prediction, the gold answer, and optionally a judged correctness flag. A *blind run* repeats inference with visual content removed and textual content unchanged. The standard local and API wrapper paths share the same stripping function; blind artifacts get a `_blind` suffix and sit beside the full run. The toolkit audit pipeline aligns frames on the dataset-level question ID and restricts them to the shared ID set.

Two rules protect validity. First, model answers are never back-filled from gold labels, whatever the judge decided; an earlier version of our own pipeline made exactly this mistake, and repairing it multiplied the unanimous-conflict count by eleven (Section 4.6). Second, predictions from which no option letter can be parsed become an abstention sentinel that is excluded from consensus votes instead of being silently counted as wrong.

3.2 Detectors

Table 1 summarizes the six detectors. Consensus, agreement, position-distribution and distractor detectors parse option letters from raw predictions and never consult a judge; visual dependency and correctness agreement consume judged labels and are re-checked under exact matching in Section 4.4.

For visual dependency, each question is categorized from model-averaged full accuracy a_q^F and blind accuracy a_q^B : *visual-dependent* ($a_q^F \geq 0.99$ and $a_q^B \leq 0.01$), *conflicting* ($a_q^B > a_q^F + \delta$, the image hurts), *visual supplement* ($a_q^F - a_q^B \geq \delta$), and *text-only* otherwise, with $\delta = 0.2$. With $n = 3$ models the accuracies live on the grid $\{0, 1/3, 2/3, 1\}$, so text-only reduces to exact equality $a_q^F = a_q^B$ and carries no tunable threshold. The visual-dependent share, in contrast, is threshold-fragile: it is 22.5% under the strict definition and 54.8% under a majority definition (Appendix B), which is why we never headline it. Consensus conflicts are tiered: *unanimous* (every parseable answer is the

Table 1: Detector taxonomy. “Judge-dep.” marks detectors whose output depends on the answer-extraction protocol.

Detector	Signal	Pathology flagged	Judge-dep.
visual_dependency	full vs. blind accuracy	no-gain-from-image items	yes
consensus_error	majority vote vs. label	label-conflict candidates	no
fleiss_kappa_agreement	inter-model κ	unstable / ambiguous items	no
correctness_agreement	judged-correctness profile	difficulty / discrimination	yes
answer_options_distribution	label-position histogram	position bias	no
distractor_similarity	pairwise option similarity	near-duplicate options	no

117 same non-gold option), *high* (all models disagree with the gold label), *medium* (a qualified majority
118 disagrees).

119 3.3 Audit statistics based on the excess over a permutation null

120 A raw zero-gain share is easy to over-read. Even if blind answers were drawn independently of the
121 items, some questions would tie full-run accuracy by chance, and that baseline moves with the pool’s
122 marginal accuracy; across our two- and three-model pools the raw share swings by 14.2 pp. The
123 layer therefore reports the share’s *excess over a permutation null*: blind score vectors are shuffled
124 across questions, with the same permutation applied to every model, which preserves each model’s
125 marginal blind accuracy and the inter-model correlation structure while destroying only the per-
126 question full-to-blind pairing. We use 999 permutations; the one-sided p -value uses the standard
127 $(g+1)/(N+1)$ correction, so the smallest reportable value is $p = 0.001$. The same module attaches
128 Wilson 95% intervals to headline proportions (item-level uncertainty), leave-one-model-out pools
129 (pool sensitivity), a split-half reproducibility check, and a stratification by official dataset categories
130 (localization and internal replication). All of it is post-processing without any model use.

131 3.4 Evaluation protocol

132 **Benchmarks.** MMStar [Chen et al., 2024]: 1,500 four-option multiple-choice items curated for
133 vision-indispensability, with six official categories of 250 items each. HallusionBench [Guan et al.,
134 2024]: 951 yes/no items built around visual illusions and language priors, used here as a deliberate
135 out-of-format probe of the detectors themselves.

136 **Models.** InternVL3-8B, Ristretto-3B and gpt-5-nano ($n = 3$): two open models and one API model.
137 This case study is sized to validate the audit machinery; leave-one-model-out ranges are reported
138 throughout and are wide (Table 2).

139 **Scoring.** The judged arm uses gpt-4o-mini as the answer-extraction judge, identical for full and
140 blind runs; the exact arm parses option letters directly from predictions with no model in the loop.
141 Consensus, κ , position and distractor detectors use the exact arm by construction. Both arms are
142 reported wherever they can differ.

143 **Reproducibility.** Every run records its git commit, command-line arguments and generation param-
144 eters in a manifest; the audit is deterministic given the artifacts and a fixed permutation seed. The
145 audit layer, the prediction artifacts for both benchmarks, and the generated reports accompany this
146 submission; Appendix A gives the exact commands that regenerate every number and figure in this
147 paper.

148 4 Results

149 4.1 A curated benchmark still carries an excess zero-gain rate

150 Figure 2 and Table 2 give the headline result. The observed zero-gain share is 20.2% (Wilson 95%
151 CI [18.2, 22.3]); the permutation null predicts 17.0%; the excess is +3.2 pp at $p = 0.001$, the floor
152 of a 999-permutation test. The excess survives every leave-one-model-out pool (+3.6 to +7.7 pp,
153 each $p = 0.001$). Across 20 random split-half checks, the median absolute difference between the

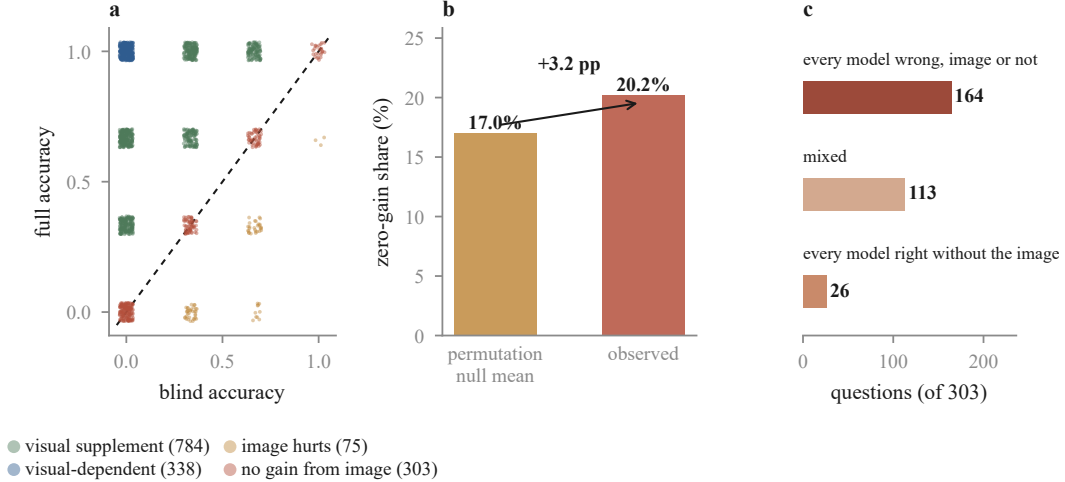


Figure 2: The zero-gain signal on MMStar. **(a)** Model-averaged full against blind accuracy per question (judged arm, jittered; three models give the 4×4 grid). Questions on the diagonal gain nothing from the image. **(b)** The observed 20.2% share against 999 item-shuffled permutations: the null centres at 17.0%, leaving a +3.2 pp excess that no permutation reaches ($p = 0.001$). **(c)** The no-gain set is heterogeneous: on 164 of 303 questions every model is wrong with and without the image, so the conservative reading is *zero aggregate accuracy gain from the image for this model pool*, and genuinely text-answerable items are only one possible component.

Table 2: Zero-gain share against the permutation null on MMStar (999 permutations, one-sided, $(g+1)/(N+1)$ correction). The excess is markedly more pool-stable than the raw share: raw spread 14.2 pp, excess spread 4.6 pp.

Model pool	Observed (%)	Null mean (%)	Excess (pp)	p
All three models	20.2	17.0	+3.2	0.001
without InternVL3-8B	27.9	24.1	+3.7	0.001
without Ristretto-3B	25.5	21.9	+3.6	0.001
without gpt-5-nano	34.4	26.7	+7.7	0.001

two half-sample excess estimates is 1.5 pp. Normalizing by the null reduces pool sensitivity: the raw share varies by 14.2 pp across pools, whereas the excess varies by 4.6 pp.

Two readings of this result must be kept apart (Figure 2c). Of the 303 zero-gain questions, 164 are items on which every model is wrong both with and without the image, which is evidence of shared failure. Only 26 are answered correctly by all three models blind. Our claim is therefore limited to equal model-averaged full and blind accuracy for this pool. The excess quantifies how often that equality occurs beyond marginal-rate arithmetic; it does not identify which underlying mechanism produced it. The modest +3.2 pp excess is also consistent with MMStar’s curation having removed most non-visual shortcuts.

4.2 The two audit signals have different category profiles

Stratifying by MMStar’s six official categories turns one number into a pattern (Figure 3). The zero-gain excess is largest in math (+6.6 pp) and science & technology (+7.5 pp), where non-visual knowledge is a plausible explanation. Coarse perception shows no positive excess (−1.7 pp). Unanimous label-conflict candidates distribute differently: they peak in the perception categories (11.6% coarse, 14.4% fine-grained) and are lowest in math and reasoning (4.4–6.8%), with science & technology intermediate (10.4%). These category profiles show that the two detectors capture distinct signals; neither profile by itself identifies the causal mechanism behind its signal.

Blind-run abstentions. The extraction discrepancy is distinct from refusal under the blind ablation. Full/blind unparseable rates are 0.0/20.5% for InternVL3-8B, 0.0/1.1% for Ristretto-3B, and

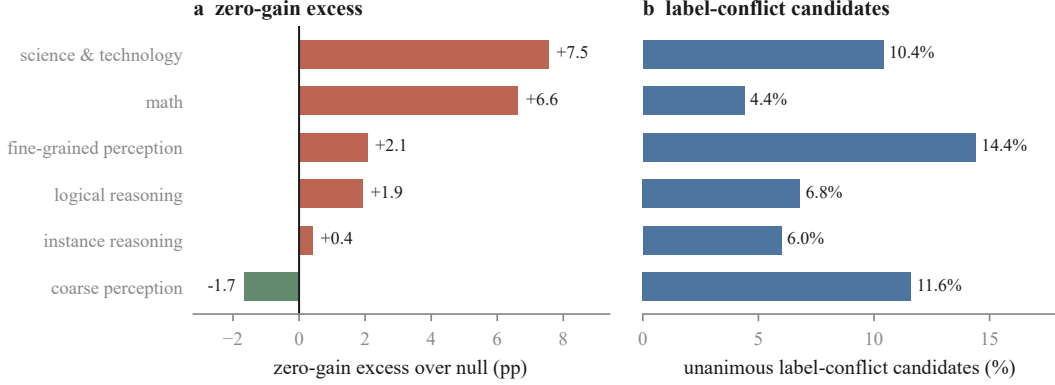


Figure 3: Per-category audit of MMStar (250 items per category), sorted by zero-gain excess. (a) The excess concentrates in mathematics and science & technology and is absent from coarse perception, a pattern consistent with greater scope for non-visual knowledge in the former categories. (b) Unanimous label-conflict candidates peak in the perception categories. The two detectors localize different candidate failure modes.

10.3/67.5% for gpt-5-nano. Consequently, zero-gain and visual-gain categories reflect both benchmark properties and model-specific willingness to answer without an image. We retain all three models in the primary pool and expose this dependence through the leave-one-model-out analysis; results should not be extrapolated to pools with different refusal policies.

4.3 Label-conflict candidates

The consensus detector flags 23.9% of MMStar items at some tier; the tier that matters is the unanimous one. On 134 questions (8.9%, Wilson 95% CI [7.6, 10.5]) every parseable model answer is the same option and it is not the gold label; on 117 of those all three models answered in parseable form, and on the remaining 17, only two models produced parseable answers because abstentions are excluded from the vote. On the 1,346 items for which all three models produced parseable option answers, complete-case Fleiss' κ is 0.544 (item-bootstrap 95% CI [0.517, 0.570]), indicating moderate agreement beyond that expected from the models' marginal answer distributions. Because the excluded 154 items are non-random unparseable outputs from gpt-5-nano, this estimate describes the complete-case subset rather than all 1,500 MMStar items. Unanimous agreement therefore stands well apart from the baseline behavior of the complete-case pool. The rate is of the same order as the annotation-error rates reported for MMStar by the independent fallibility analysis of Li et al. [2025], which did collect fresh human labels.

We deliberately call these *candidates*, and the reason is measurable in our own pool. Errors of different models are correlated [Kim et al., 2025]: when two of our models are both wrong, they pick the same wrong option in 64–74% of cases, against 26–27% when the item-to-item pairing is destroyed by the same shuffle that defines our null. Unanimity therefore cannot adjudicate a gold label on its own. It concentrates human review, which remains the arbiter. The audit layer ships the flagged list with per-question model votes for exactly that purpose; a human verification campaign over a stratified sample is the natural next step and sits outside this paper's claims.

4.4 Removing the LLM judge changes almost nothing, for a reason

Audit claims should not silently inherit an LLM judge [Rosenthal et al., 2025]. Four of six detectors never use one (Table 1). For the two that do, we recomputed everything under exact letter matching on identical predictions (Figure 4a): category shares move by at most 2.1 pp (zero-gain 19.8% against 20.2%; visual-dependent 20.5% against 22.5%; supplement 53.6% against 52.3%; conflicting 6.1% against 5.0%), and nothing in Sections 4.1–4.3 changes.

The instructive part is where the two arms disagree. 122 of 1,500 items (8.1%) change visual-dependency category between arms, and every one of them traces to gpt-5-nano: the judge and the letter parser score that model differently on 229 model-item pairs, and the two open models on none

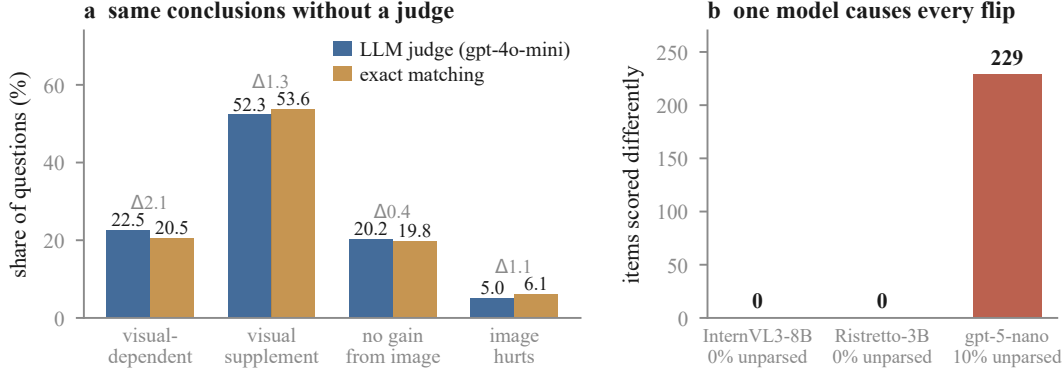


Figure 4: Judge dependence of the audit on MMStar. **(a)** Category shares under the gpt-4o-mini judge and under exact letter matching, on identical predictions: every share moves by at most 2.1 pp and no conclusion changes. **(b)** The disagreement has a single source: for the two open models the judge and the letter parser score every item identically, while gpt-5-nano, whose answers are unparseable on 10% of items, accounts for all of it.

at all (Figure 4b). The mechanism is prosaic: gpt-5-nano answers in free-form prose on 10% of items, which a regular expression cannot read but a judge can. The judge is thus a robustness arm: for models with canonical outputs exact matching suffices, and for models without canonical outputs the audit report can say so per model instead of averaging the discrepancy away.

4.5 Where the detectors stop working: a binary-format probe

We ran the same audit on HallusionBench [Guan et al., 2024], whose 951 items are yes/no questions, to measure how the detectors behave outside their design format. With two answer options blind guessing sits at 50%, unanimity is easy to obtain by chance, and the multiple-choice consensus machinery does not apply at all: option letters do not parse and the exact arm reports abstention everywhere. The zero-gain excess is still large and significant (+5.4 pp over an 18.5% null, $p = 0.001$), but the decisive calibration test fails. Ranking items by their continuous visual gain recovers HallusionBench’s own visual-dependent against visual-supplement design labels at AUC 0.49, that is, at chance (Appendix C). One directional signal survives and matches the benchmark’s intent: the excess is approximately 2.2 times larger on design-VS items (+7.6 pp) than on design-VD items (+3.4 pp). We conclude that the current evidence supports use on the four-option MMStar setting evaluated here and places binary formats outside the demonstrated scope. Calibration across other k -way multiple-choice benchmarks remains to be measured.

4.6 Auditing the audit

Audit pipelines can manufacture their own findings, so we checked ours against the known failure modes. *No-op perturbation*: the blind run strips images centrally, blind transcripts contain model statements that no image is present, and blind accuracy falls towards the option prior in the perception categories. *Extraction artifacts*: the judged-against-exact arm (Section 4.4) measures precisely this and attributes all discrepancy to one model. *Unfaithful scoring*: the judged arm is VLMEvalKit’s native protocol, unmodified. *Broken pairing in uncertainty*: the permutation null is item-paired by construction, and only the full-to-blind linkage is destroyed. *One scalar, two readings*: every headline is disaggregated (category strata, per-pool tables, and the 164-of-303 composition in Figure 2c). One bug we did catch and fix is reported in Section 3.1: back-filling gold labels into model answers had suppressed unanimous conflicts from 134 to 12, and the layer now refuses to do it.

5 Limitations

(i) Three models. Consensus unanimity among correlated models [Kim et al., 2025] is weak per-item evidence (we measure 64–74% same-wrong-option agreement), so the detector produces candidates for human review, and the leave-one-model-out ranges in Table 2 show how much three models leave

open. (ii) The label-conflict candidates in this paper are unverified by humans; the precision of the consensus detector is therefore unmeasured, and the 8.9% figure is a candidate rate. (iii) Unparseable predictions become misses for correctness-based categories and abstentions for answer-consensus statistics. Blind unparseability reaches 67.5% for one model, so the measured gain partly reflects model-specific refusal policy. (iv) Equality of model-averaged full and blind accuracy is a zero aggregate gain; the same models or answers need not remain correct. It may combine language priors, answer-option statistics, training-data contamination and offsetting per-model transitions; the audit does not separate these mechanisms (cf. Song et al., 2024). (v) The visual-dependent and supplement categories depend on the threshold δ ; we headline only threshold-free quantities and publish the sweep (Appendix B). (vi) Binary yes/no formats are out of calibration, as measured in Section 4.5. (vii) One benchmark is audited in depth; the claims are about the audit machinery, and its base rates should not be extrapolated to other suites without running it.

6 Conclusion

An evaluation toolkit already stores what a benchmark audit needs. Adding one text-only rerun and a statistics pass turns those artifacts into per-item findings with uncertainty attached: on MMStar, a +3.2 pp excess zero-gain rate concentrated in knowledge-heavy categories, and 134 unanimous label-conflict candidates concentrated in perception categories. The former is an operational statement about aggregate accuracy and leaves the underlying mechanism open; the latter prioritizes items for human review. These findings show why benchmark scores should be accompanied by audit statistics, model-pool sensitivity and refusal profiles. We are extending the pool to six models, running the human verification the consensus detector calls for, and sweeping the remaining VLMEvalKit benchmarks; the layer, artifacts and reports are released so that others can do the same.

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A Reproducing every number in this paper

The audit consumes only the per-model result files that VLMEvalKit writes for a normal evaluation, plus the matching blind-run files. For each benchmark, the following two commands regenerate its statistics and figures; the HallusionBench invocation additionally passes `-design-prefix VD` to both scripts.

```
python scripts/audit_significance.py \
  -full <model>_MMStar_gpt-4o-mini_result.xlsx (one per model) \
  -blind <model>_MMStar_gpt-4o-mini_result_blind.xlsx \
  -models InternVL3-8B Ristretto-3B gpt-5-nano \
  -strata-column category -perms 999 -seed 7 -out <report dir>

python scripts/audit_paper_figures.py \
  -full ... -blind ... -models ... -stats-json ... -out
paper/figures
```

The blind artifacts themselves come from a normal evaluation invocation with the `-blind` flag, which is the only step that requires inference:

```
python run.py -data MMStar -model <model> -blind
```

The statistics script emits the canonical JSON report. The figure script reads all reported aggregate and stochastic values from that JSON and uses the evaluation files only for deterministic per-item panels; it does not rerun the permutation null. The scripts depend only on pandas/openpyxl (plus matplotlib for figures) and do not import the toolkit itself. Randomness enters only when the canonical report is generated, through the recorded permutation seed.

Answer extraction (exact arm). A prediction is mapped to an option letter by three ordered rules: a bare letter, possibly punctuated; a letter followed by a delimiter at the start of the string; or a letter following the words “answer” or “option”. Anything else becomes the abstention sentinel Z, which is excluded from consensus votes. This mirrors the toolkit’s own extractor, so the audit does not introduce a second, divergent notion of what a model answered.

Permutation null. For each of 999 permutations, one random permutation of question IDs is drawn and applied to the blind hit vectors of *all* models simultaneously. This preserves each model’s marginal blind accuracy and the correlation between models’ blind answers, and destroys only the per-question pairing between full and blind runs. The zero-gain share is recomputed on each permuted dataset with the identical category rule; the reported excess is observed minus null mean, and the one-sided p -value is $(g+1)/(N+1)$, where g counts permutations with a share at least as large as observed. Stratified nulls repeat the procedure within each category. For the split-half check, questions are randomly halved and the excess is computed in each half. Across 20 resplits, the median absolute difference between half-sample estimates is 1.5 pp on MMStar and 2.1 pp on HallusionBench; these values are stability diagnostics.

B Threshold sensitivity of the visual-dependent share

The visual-dependent share depends strongly on its definition, which is why the paper never head-lines it. On MMStar (judged arm), the strict all-or-none definition flags 338 items (22.5%), while a majority definition (at least two of three models correct with the image, at most one without) flags 822 items (54.8%). On HallusionBench the same pair is 140 (14.7%) against 478 (50.3%). The text-only category has no threshold at $n = 3$, since the accuracy grid makes it exact equality, and the unanimous consensus count has none at any n ; those are the quantities the paper reports.

C Binary-format calibration check

Design labels are read from the question index prefix that HallusionBench itself assigns (VD for visual-dependent items, the remainder treated as visual-supplement), giving 591 positives and 360 negatives. The AUC is computed on the continuous per-question visual gain $a_q^F - a_q^B$, with ties

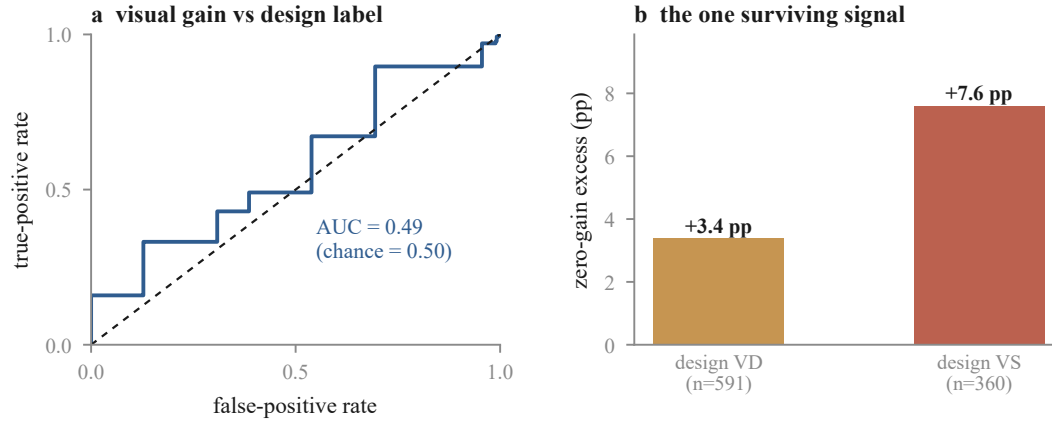


Figure 5: HallusionBench (951 yes/no items, same three models). **(a)** Ranking items by the continuous visual gain does not recover the benchmark’s own design labels (AUC 0.49 against 0.50 for chance). **(b)** The one directional signal that survives is consistent with the benchmark’s design: zero-gain excess is approximately 2.2 times larger on visual-supplement items than on visual-dependent items.

384 counted as half wins. Together with the degenerate letter-extraction behaviour described in Sec-
 385 tion 4.5, this is the evidence for excluding binary formats from the detectors’ current calibration.